

Some notes of Microsoft Exchange Deserialization RCE (CVE-2021-42321)

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How should I begin, ...

About a month ago, everyone in the researcher/APT/blue-team world turned their eyes to China, when VMWare, Exchange, iPhone ... were pwned one after another at the Tianfu Cup.

For me, the notable thing this time is that Exchange got pwned again after half a year of silence.

Rather unlucky for the Exchange dev team, who all year round only sit around eating and fixing bugs, and the appearance of the most recent vulnerability also owes something to them ...

This time I am no longer that excited when MS releases a patch for Exchange, because I am quite busy and no longer want to burn more than half of my month sitting and diffing patches. (Even so, curiosity did not let me sit still.)

Yesterday the blog of organization X also published an analysis of this bug, but it is perhaps still missing something, so I had to set about writing this article, in the form of a note that I may need to refer back to later without missing important information. Let us get started.

...

From MS's Advisories, we can see that this is a post-auth vulnerability:

And there is a patch for exactly these 4 versions of Exchange:

The first thing that puzzles me here is that an RCE bug like this, shown off at the Tianfu Cup, ought to be Pre-Auth, right ☐. Perhaps MS has hidden the authentication bypass bug somewhere, and we will have to wait for the December or January 2022 patch to know how it really is.

The version I used for the lab this time is Exchange 2019 CU9, and only when upgrading to CU10 did I discover where the bug was ☐.

While looking for a patch to diff, I discovered that on 11/10/2021 Microsoft had tried to patch something (just a few days before TianfuCup)

To get a broader view, I chose the July patch to diff; if the bug was brought out and shown at Tianfu, that means it must have been found a few months earlier.

After decompiling + removing the junk and then diffing, only about 270 files really had changes (J°☐°☐☐☐ LL).

Almost all the changes in this patch are meant to prevent Insecure Deserialization somewhere (where exactly is still unknown).

#THE SINK

After hanging around this pile of patches for a few days, I discovered that something was off in the class **TypedBinaryFormatter** (this class was removed in the new patch).

The *Deserialize()* method of this class takes a parameter "**SerializationBinder**", however the handling code inside does not use this "**binder**" at all:

This method calls **ExchangeBinaryFormatterFactory**.*CreateBinaryFormatter()* *to create an instance of BinaryFormatter for the deserialization:

The *CreateBinaryFormatter()* **method simply creates an instance of **BinaryFormatter** **with **Binder=ChainedSerializationBinder**.**

With the parameters passed in from **TypedBinaryFormatter.Deserialize()**, we get a ***ChainedSerializationBinder* ***with:*

- strictMode = **false**
- allowList = System.DelegateSerializationHolder
- allowedGenerics = null

That means the **binder** ***passed in from **TypedBinaryFormatter.Deserialize()** *has no effect whatsoever.*

Continuing to look into the method **ChainedSerializationBinder.BindToType()**, this method goes on to call *ValidateTypeToDeserialize()* for the main handling.

And the way the *ValidateTypeToDeserialize()* method handles things also has a few rather interesting (hard to understand ?!) points

The logic of this check is as follows:

Branch **(1)**:

- If strictMode = **false**, the current class is not in the allow list, and this class is in the blacklist => throw **InvalidOperationException**.
- Although this piece of code sits inside a try catch statement, the catch part does not catch **InvalidOperationException** ***but only catches **BlockedDeserializationException**.*
- Therefore, if branch **(1)** throws **InvalidOperationException**, the handling is effectively finished, and the data will no longer be deserialized

Looking more closely at the blacklist of **ChainedSerializationBinder**, this blacklist is built by the method *BuildDisallowedTypesForDeserialization()*:

This blacklist includes the common gadget chains, however it seems there is a small typo here:

The correct class must be System.Security.**Claims.ClaimsPrincipal**

And in this blacklist I also see the absence of quite a few common gadget chains, notably **TypeConfuseDelegate** (fact: even in the new patch the ***SortedSet* ***class of this gadget*

chain is still not blocked (ಠ_ಠ)).

So it is entirely possible to use gadget chains to bypass it; the handling of branch **(1)** of ****ChainedSerializationBinder**** is completely useless.

Continuing with branch (2):

If the Class does not satisfy the condition (blacklisted, not in the whitelist ...) then a ****BlockedDeserializeException**** will be thrown ...

However, right in branch 2 below it, it is caught very neatly, and only throws when **flag** = true. With flag here = strictMode, the thing that is set to false by default

Summing up both things, we get a useless **ChainedSerializationBinder**)))

That gives us an Insecure Deserialization vulnerability right at

TypedBinaryFormatter.Deserialize(), the thing that was born to make Deserialize safer ☐

...

#THE SOURCES

As usual, I traced back to the methods that call

TypedBinaryFormatter.DeserializeObject()

This tracing takes a bit of time, because dnspy is dumb at finding usages through an interface, so I had to do this by hand! (Or you can use JetBrains Rider to find the Hierarchy)

Among them, **OrgExtensionSerializer.TryDeserialize()** is calling the method

ClientExtensionCollectionFormatter.Deserialize()

This method takes the stream from **UserConfiguration.GetStream()** and then passes it into *Deserialize()*

After googling for a while, I learned that this UserConfiguration can be created even by an ordinary user (<https://docs.microsoft.com/en-us/exchange/client-developer/web-service-reference/createuserconfiguration-operation>).

And while tracing the code, I discovered that ****CreateUserConfiguration**** also has a BinaryData field (not mentioned in MS's docs).

When calling **CreateUserConfiguration.Execute()**, the method

UserConfigurationCommandBase.SetProperties() is also called along with it. This method calls *SetDictionary()*, *SetXmlStream()* and *SetStream()*.

In it, the *SetStream()* method takes the data from the BinaryData field, then Base64 decodes it and saves it as a stream:

=> That means we can fully control the value of **UserConfiguration.*GetStream()** **without needing any special permission.*

To control this data, the request to EWS is simply as follows:

In it, CfgName, folder id and change key need to be adjusted appropriately for the poc to work

...

Continue tracing back through the code until we reach the class **GetClientAccessToken**

After a round of googling, it turns out this is a feature that can be called from EWS (<https://docs.microsoft.com/en-us/exchange/client-developer/web-service-reference/getclientaccess-token-operation>), a sample request looks like this:

...

At this point reaching the Sink can be considered complete, and it can be summarized in 2 steps as follows:

- **Step 1:** Create a UserConfiguration containing BinaryData = GadgetChain
- **Step 2:** Call GetClientAccessToken to trigger Deserialization

Just use the TypeConfuseDelegate gadget chain and pop calc.exe ...

In summary, we have the following stack trace when exploiting:

...

#THE IMPROVEMENT

However, that is only to show off in the lab; in reality it will be like this:

And in reality, all processes spawned from w3wp.exe get very special attention from AV/EDR/sysmon ... If you can RCE, either it is in a lab, or, second, congratulations, you may well have fallen into someone's honeypot (°_°)

In the current case, we can use the gadget chain **TypeConfuseDelegate**, or another gadget, **ClaimsPrincipal**.

With the current design of **TypeConfuseDelegate**, we can only RCE from it (other things are possible but will be mentioned later), while **ClaimsPrincipal* has a method **OnDeserializedMethod()*, which is called when Deserialize succeeds. This method in turn calls *DeserializIdentities()* to restore the Identity field of this object:

DeserializIdentities() in turn calls **BinaryFormatter.Deserialize()**, which is what causes Second-Order Deserialization, as has been written very clearly here (<http://muffsec.com/blog/finding-a-new-datacontractserializer-rce-gadget-chain/>):

These facts allow us to wrap a more complex gadget chain — the thing that was filtered before, for example wrapping the gadget "**ActivitySurrogateSelector**" which allows loading a dll:

<illustration image>

The idea of loading a dll was taken from this blog post: http://www.zcgonvh.com/post/analysis_of_CVE-2020-17144_and_to_weaponizing.html

The bug analyzed in that article is also quite similar to this Exchange 2019 bug; in the article, the author uses the gadget chain "**ActivitySurrogateSelector**" to load a dll file, allowing a web shell to be injected into memory:

This idea is quite good; I also broadened my mind quite a lot while reading this blog.

And to apply it to the current context, instead of calling **Process.Start()**, I switched to using **Microsoft.JScript.Eval.*JScriptEvaluate()** to eval JScript, an approach similar to

chopper shell or antsword.

Being able to execute JScript will make us more comfortable in reading files and executing commands while being hard to detect.

This is the piece of code used to replace **Process.Start()**

One small note: because we cannot access the page context of IIS, we will not be able to call methods such as: **Response**.Write, System.Thread ... But here this much is enough!

The next thing to do is to copy the DLL file just compiled into the same folder as ysoserial, then wrap the **ActivitySurrogateSelector** Object inside the gadget chain **ClaimsPrincipal** as below (ysoserial currently does not have the ClaimsPrincipal gadget, however it is quite similar to ClaimsIdentity so you can copy paste the code and modify it a little and it will work):

- the generated payload will be very long

...

Memory shell, the payload is fully prepared, however there is still a very important problem here:

Recent .NET versions have implemented some checking mechanisms to prevent abuse of the gadget chain **ActivitySurrogateSelector**, which is also why in ysoserial you can see a gadget chain used to disable this checking mechanism:

This gadget chain simply uses the gadget **TextFormattingRunProperties** *to call methods that change the value of the property*

"microsoft:WorkflowComponentModel:DisableActivitySurrogateSelectorTypeCheck"

TextFormatting *is also in the blacklist of ChainedSerializationBinder*, so it must be wrapped in **ClaimsPrincipal** to be executed!

=)))) I've been laughing for hours while making this meme

And this is the result when using this gadget chain:

The gadget chain returns an **InvalidCastException** and then the process **w3wp.exe** crashes! This has also been mentioned at <https://www.netSPI.com/blog/technical/adversary-simulation/re-animating-activitiesurrogateselector/>

This gadget chain works as follows: when deserialization is done the program goes on to call the constructor, and in the constructor of this class it calls

GetObjectFromSerializationInfo() and then goes on to call **XamlReader.Parse()**:

In it, **XamlReader.Parse()** is exactly the method that handles and executes the commands in the xaml payload that is passed in.

After executing, the constructor of **TextFormattingRunProperties** *casts to the type "Brush", however our data does not satisfy that, so it causes an InvalidCastException*, and the program does not catch this exception either, which is why the whole w3wp.exe process crashes.

So it is not possible to use **TextFormattingRunProperties** *to Parse the Xaml Payload and disable the check. We need to find another gadget, another combination*

between gadgets that can disable that check, or find another gadget to call `**XamlReader.Parse()`.

After going around in circles for a while, I took another look at `TypeConfuseDelegate`; this gadget chain looks quite simple, and is not hard to modify either:

The key point of this gadget is the part that modifies `invoke_list`, which allows passing in a static method that has been packaged; roughly speaking, we can call a static method other than **`Process.Start()`**.

(◕‿◕)

Combined with **`XamlReader.Parse()`** also being a static method, everything becomes much simpler. The way to modify it for the combination is also just that simple:

After combining it with the xaml payload that disables the check, everything worked nicely; how to use `Eval JScript` next is something I may talk about in another episode!

PoC video: <https://www.youtube.com/watch?v=Fmx6JISABAQ>

PoC: I don't think i will ...

Checking all the current versions of Exchange, I determined that this PoC only works on the following versions:

- Microsoft Exchange 2019 CU10, 11
- Microsoft Exchange 2016 CU21, 22

This vulnerability may have happened because the developer team got confused when migrating to the new way of using **`ChainedSerializationBinder`**, or for some other reason ... `¯\_(ツ)_/¯` who know.

Thank you all for your interest and for following along

Thank @peterjson for collaborating and English version of this blog post, and an anonymous man for helping!

Jang

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